

SALES PERFORMANCE ANALYSIS

Using Microsoft Excel Advanced Functions and Tools

Project Documentation

Submitted as part of the Excel Advanced Functions and Tools training module

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1. Introduction and Objective

This document presents the methodology, analysis, and findings of a Sales Performance Analysis project carried out in Microsoft Excel. The project applies a set of advanced Excel functions and tools — including text functions, lookup formulas, conditional aggregation, PivotTables, and Pivot Charts — to a raw transactional sales dataset, with the goal of converting unstructured, inconsistent data into a clean, analysis-ready format and extracting actionable business insight from it.

The specific objectives of the project were:

- To clean and standardize a raw sales dataset containing inconsistent region and product category labels.
- To calculate descriptive statistics such as average sales, average profit, and average discount using conditional formulas.
- To summarize total sales and profit across regions and product categories using PivotTables.
- To visualize regional and category-level performance using Pivot Charts.
- To identify the best- and worst-performing regions and product categories by sales and profitability.

2. Dataset Description

The analysis is based on a transactional sales dataset (“Raw data” sheet) consisting of 4,000 individual order records spanning the period from 16 November 2022 to 15 November 2024. Each record captures a single sales transaction across four geographic regions and five product categories.

2.1 Data Fields

Field	Description
Order ID	Unique alphanumeric identifier for each transaction
Order Date	Date the order was placed (Nov 2022 – Nov 2024)
Region	Sales region: East, West, North, South
Product Category	Category of product sold: Clothing, Electronics, Furniture, Groceries, Sports
Sales Amount	Total value of the order, in currency units
Quantity Sold	Number of units sold in the order
Discount (%)	Discount percentage applied to the order
Profit	Profit earned on the order, in currency units

2.2 Data Quality Issues Identified

Initial inspection of the raw data revealed significant inconsistency in two categorical fields — Region and Product Category — introduced through inconsistent capitalization, stray symbols, extra spaces, and character substitutions (e.g. “\$” in place of “S”). This is a common simulation of real-world “dirty data” and made standardization a necessary first step before any aggregation could be trusted.

Field	Distinct Raw Values Found	Clean Values After Standardization
Region	25 (e.g. “norTh”, “Mes\$t”, “\$0uTH”, “N o r t H”, “Northhhhh”)	4 — East, West, North, South
Product Category	28 (e.g. “Cloth ing”, “FURNiture”, “\$port\$”, “Growcreie\$”)	5 — Clothing, Electronics, Furniture, Groceries, Sports

Standardizing these fields was essential: without cleaning, PivotTable summaries would have split each true category into a dozen or more spurious sub-groups, severely understating totals for any single, correctly-spelled label.

3. Methodology

The project was carried out in a sequence of tasks, each applying a specific Excel function or tool. The table below summarizes the techniques used and their purpose in this project.

Task	Technique Applied	Purpose in This Project
Task 1–2	UPPER, PROPER text functions	Standardized inconsistent text casing in Order ID and Region/Category fields as a first cleaning pass
Task 3	VLOOKUP, AVERAGEIF	Looked up region names against a clean reference table and calculated average sales per region
Task 4	SUMIF, AVERAGEIF	Calculated total sales per region and average profit/discount specifically for the Furniture category
Task 5	PivotTable (Region × Category)	Built a two-level summary of Sum of Sales Amount and Sum of Profit, broken down by Region and then Product Category
Task 8	PivotTable (Region → Category)	Produced a region-led summary table of total sales drilled down by category
Task 9	GETPIVOTDATA	Extracted specific summary figures (e.g. total profit, total sales) directly from existing PivotTables for use elsewhere in the workbook
Task 10	Conditional Formatting(sales>4000)	To automatically highlight or format sales above 4000 in spreadsheet programs like Microsoft Excel or Google Sheets , you can use a Greater Than rule. This will dynamically apply a specific style (such as a green background) to any cell where the value exceeds 4000
WK Sheet	PivotTable (Category totals)	Summarized grand total sales by product category across all regions combined

3.1 Data Cleaning

The Order ID, Region, and Product Category fields were passed through UPPER() and PROPER() functions to normalize text casing as an initial cleaning step. For the purposes of grouping and aggregation, Region and Product Category values were treated as their standardized equivalents (East/West/North/South and Clothing/Electronics/Furniture/Groceries/Sports respectively) throughout the rest of the analysis.

3.2 Lookup and Conditional Aggregation

A small reference table of region names was created, and VLOOKUP was used to match each transaction's region against it. AVERAGEIF and SUMIF were then applied to calculate:

- Average Sales Amount per region
- Total Sales Amount per region
- Average Profit specifically for the Furniture category
- Average Discount percentage specifically for the Furniture category

=AVERAGEIF(\$C\$2:\$C\$4001, K3, \$E\$2:\$E\$4001)	→ Average sales by region
=SUMIF(\$C:\$C, J3, \$E:\$E)	→ Total sales by region
=AVERAGEIF(D:D, "furniture", H:H)	→ Average profit for Furniture
=AVERAGEIF(D:D, "furniture", G:G)	→ Average discount % for Furniture

3.3 PivotTable Summarization

PivotTables were built with Region and Product Category in the Rows area and Sum of Sales Amount / Sum of Profit in the Values area. This produced a hierarchical summary showing total sales and profit at both the region level and the region-category intersection level, fully recalculating from the cleaned source data.

3.4 GETPIVOTDATA for Cross-Sheet Reporting

GETPIVOTDATA was used to pull specific aggregate figures (such as grand total Sales Amount and grand total Profit) directly out of the PivotTables and reference them in other parts of the workbook, avoiding duplicate manual entry of summary figures.

3.5 Visualization with PivotCharts

Three PivotCharts were created from the summarized data:

1. A horizontal bar chart comparing total Sales Amount across the four regions.
2. A pie chart showing the share of total sales contributed by each of the five product categories.
3. A clustered bar chart breaking down sales by product category within each region, supported by slicers for interactive filtering by region and category.

4. Findings

4.1 Total Sales and Profit by Region

Across all 4,000 transactions, total Sales Amount reached 10,109,783.69, generating total Profit of 4,030,895.69 — an overall profit margin of approximately 39.9%. The East region recorded the highest total sales, narrowly ahead of North, South, and West, with all four regions clustered within roughly 4.6% of one another, indicating a fairly even geographic distribution of sales volume.

Region	Total Sales Amount	Total Profit	Profit Margin
East	2,595,496.48	1,050,859.93	40.5%
North	2,539,669.00	970,306.07	38.2%
South	2,494,411.28	1,010,902.55	40.5%
West	2,480,206.93	998,827.14	40.3%
Grand Total	10,109,783.69	4,030,895.69	39.9%

Table 4.1 — Total Sales Amount and Profit by Region (PivotTable, Task 5 / Task 8)

4.2 Sales Share by Product Category

Clothing was the leading product category by total sales, contributing 21.3% of overall revenue, followed closely by Furniture and Electronics, each at approximately 20.3%. Groceries was the lowest-performing category at 18.4% of total sales. The relatively narrow spread (18.4%–21.3%) suggests a well-diversified product mix with no single category dominating overall performance.

Product Category	Total Sales Amount	Share of Total
Clothing	2,156,007.58	21.3%
Furniture	2,049,496.16	20.3%
Electronics	2,047,834.93	20.3%
Sports	1,992,219.96	19.7%
Groceries	1,864,225.06	18.4%
Grand Total	10,109,783.69	100.0%

Table 4.2 — Total Sales Amount by Product Category (PivotTable, WK Sheet)

4.3 Regional Breakdown by Product Category

Combining region and category reveals more pronounced variation than either dimension alone. Sports was the strongest category within the East region (575,612.96), while Clothing led in the North (575,975.88) and South (529,784.90), and Electronics led in the West (574,757.97). Groceries was consistently the weakest or near-weakest category in every region, reinforcing the category-level finding above.

Region	Category	Sales Amount	Profit
East	Clothing	558,307.99	217,360.53
East	Electronics	453,722.85	194,570.42
East	Furniture	558,116.90	227,380.53
East	Groceries	449,735.78	181,843.68
East	Sports	575,612.96	229,704.77
North	Clothing	575,975.88	207,145.82
North	Electronics	527,159.55	203,772.53
North	Furniture	494,993.63	209,391.78
North	Groceries	471,913.51	172,529.97
North	Sports	469,626.43	177,465.97
South	Clothing	529,784.90	224,799.76
South	Electronics	492,194.56	188,923.50
South	Furniture	487,656.54	189,883.78
South	Groceries	496,331.25	211,125.65
South	Sports	488,444.03	196,169.86
West	Clothing	491,938.81	206,240.56
West	Electronics	574,757.97	216,391.51
West	Furniture	508,729.09	208,424.23
West	Groceries	446,244.52	174,106.08
West	Sports	458,536.54	193,664.76

Table 4.3 — Sales Amount and Profit by Region and Product Category (PivotTable, Task 5)

4.4 Average Sales by Region

Average order value (Sales Amount per transaction) was highest in the North region at 2,573.12 and lowest in the South at 2,494.41, a spread of roughly 3%. This indicates that while South generates slightly lower total sales than East, this is driven more by transaction volume and mix than by materially smaller order sizes.

Region	Average Sales Amount per Order
East	2,507.73
North	2,573.12
South	2,494.41
West	2,535.99

Table 4.4 — Average Sales Amount by Region (AVERAGEIF, Task 3)

4.5 Furniture Category Deep Dive

Targeted conditional analysis of the Furniture category (Task 4) found an average profit per order of 1,023.38 and an average discount of 25.2%. Comparing this to the category's overall total profit share (20.3% of total sales) suggests Furniture sustains healthy per-order profitability eve at a discount level close to the dataset-wide norm.

5. Visualisation

Three Pivot Charts were produced directly from the PivotTable summaries described in Section 4, allowing the underlying figures to be interpreted at a glance:

- ✓ Bar Chart — Total Sales Amount by Region: confirms the close competition between East, North, South, and West, with no region varying by more than roughly 5% from the others.
- ✓ Pie Chart — Sales Share by Product Category: visually confirms the balanced product mix, with Clothing as the largest slice and Groceries the smallest.

Clustered Bar Chart — Sales by Category within each Region (with Slicers): enables interactive drill-down, letting a user filter to a single region or category and immediately see how it compares across the other dimension

6. Conditional Formatting

Rule Logic: Cell Value > 4000

Short Description: Highlights high-performing sales figures exceeding 4000 in green.

Business Purpose: Used to quickly identify and evaluate targets or transactions that meet the 4000 qualification threshold.

7. Conclusion

This project demonstrates an end-to-end Excel-based sales analysis workflow: starting from a deliberately messy 4,000-row transactional dataset, applying text functions to standardize categorical fields, using lookup and conditional aggregation functions (VLOOKUP, SUMIF, AVERAGEIF) to derive regional and category-level statistics, and finally summarizing and visualizing the results using PivotTables, GETPIVOTDATA, and PivotCharts with slicers.

The resulting analysis shows that sales performance is fairly evenly distributed across all four regions (within a ~5% band) and across all five product categories (within a ~3-percentage-point band of sales share), with Clothing as the top category overall and Groceries consistently the weakest performer across every region. The Furniture category, while only mid-pack in total sales, shows a strong average profit per order even under a relatively high average discount rate.

From a methodological standpoint, the project illustrates why data cleaning is a prerequisite for reliable PivotTable analysis: had the 25 inconsistent Region labels and 28 inconsistent Category labels not been standardized, the summary tables in Section 4 would have fragmented into dozens of near-duplicate rows, materially understating true totals for every group.